Pseudocode File

Emily Machel

```
1)

# Begin
FUNCTION solution(A,k)
    IF A is empty
        RETURN A
    SET result TO a copy of A
    REPEAT k times
        SET last TO remove the last element of result
        INSERT last @ the front of result
    RETURN result
# End
```

2)

```
#Begin
FUNCTION solution(A)
    SET counts TO an empty dictionary
    FOR each number in A
        IF number is in counts
            SET counts[number] TO counts[number] + 1
        ELSE
            SET counts[number] TO 1
    FOR each number, count in counts
        IF count % 2 == 1
            RETURN number
#End
```

3)

```
#Begin
FUNCTION solution(X, Y, D)
    SET distance_across_road TO Y - X
    SET jumps TO distance_across_road // D
    IF distance_across_road % D > 0
        SET jumps TO jumps + 1
    RETURN jumps
#End
```

4)

```
#Begin
FUNCTION solution(A)
   SET N TO length of A
   SET expected_sum TO (N+1)*(N+2) // 2
   SET actual_sum TO sum of all elements in A
   SET missing_number TO expected_sum - actual_sum
    RETURN missing_number
#End
```

5)

```
#Begin
FUNCTION solution(A)
    SET total TO sum of A
    SET left TO 0
    SET min_diff TO None
    FOR P from 1 to N-1
        SET left TO left + A[P-1]
        SET right TO total - left
        SET difference TO absolute value of (left - right)
        If min_diff is None OR difference < min_diff
            SET min_diff TO difference
    RETURN min_diff
#End
```